

Trace Analysis Report — Inventory (EWM) Load

SAP S/4HANA — Transaction /SCWM/ISU

Date: 2026-08-01 **Project:** ConcentoRDG **Trace source:** c:\Codes\Paralelismo\TraceLog Inventory\

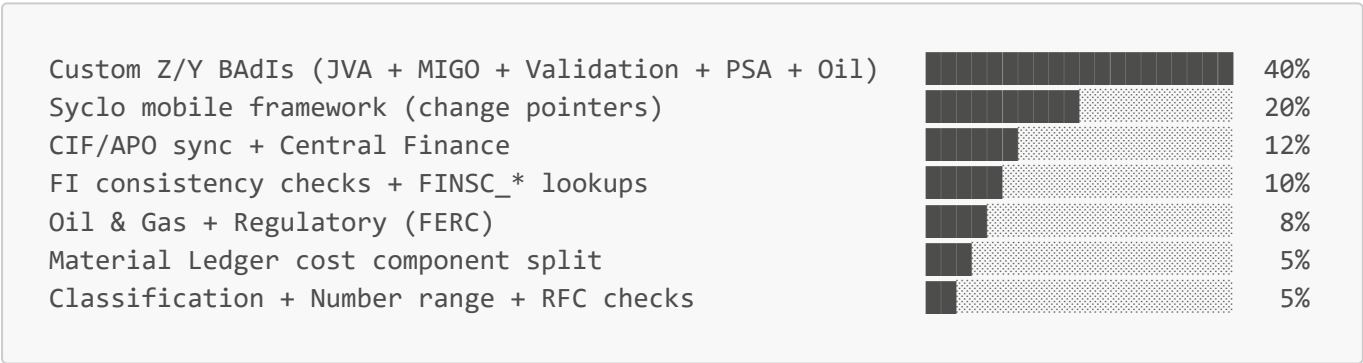
- `DataAccessExternal.csv`
- `DataAccessInternal.csv`
- `DBTables.csv` **Analyzed object:** Inventory (EWM) — WRICEF ZCON1091 **Current load method:** T-code `/SCWM/ISU` (program `/SCWM/R_INITIALSTOCKUPLOAD`) **Current execution time:** ~24 hours for 28,687 records (~0.33 records/second)

1. Executive Summary

The `/SCWM/ISU` transaction itself is **not** the bottleneck of the 24-hour load window.

The trace shows that the real cost per record comes from **custom Z/Y BAdIs, industry solution integrations, and configuration lookups** that fire on every goods movement generated by the Initial Stock Upload.

Estimated technical breakdown of the cost per record:



Consequence: the biggest gain will come from **reducing what fires during the cutover window**, not from replacing `/SCWM/ISU`.

Realistic reduction potential: from ~24h to **~6–14h**, depending on how many optimization levels are applied.

2. Trace Findings — Top Offenders

2.1 Custom BAdIs and User Exits (Z / Y)

Custom classes appearing repeatedly across the trace, called per goods movement:

Custom class	Purpose	Trace evidence
<code>ZCL_JVA_FI_DOC_DFTCO_IMP</code>	Default Cost Object for Joint Venture Accounting	20+ hits on USOBX_C, ZI_WRAP_SETLEAF, ZI_JNTVNTRSUBSTITUTECOSTOBJECT

Custom class	Purpose	Trace evidence
ZCL_MIGO_BADI_HT_MODIFY	Custom MIGO — Handling Type modification	USOBX_C, ZI_CONSTANTPARAMETERSF
ZCL_DEFAULT_HT_GOODSMVT_CREATE	Custom Goods Movement default	UST12, ZI_CONSTANTPARAMETERSF, I_PRODUCTPLANTBASIC
ZCL_OIL_MATERIAL_PROCESS	Oil & Gas material processing	USOBX_C, I_PRODUCT
YCL_MB_MIGO_BADI_LOGIC	Extra MIGO logic	YI_ZERO_PRICE_MAINT, YI_PLANT_NODETYPE_RESTRICTION
YCL_MB_DOCUMENT_BADI	Document-level BAdI	ZI_CONSTANTPARAMETERSF, IPRODUCT
ZCL_VALIDATE_MATDOC	Material document validation	IPRODUCT
ZCL_EXP_INV_OPEN_BIN_BTE	Business Transaction Event (BTE) — open bin	ZI_CONSTANTPARAMETERSF
ZCL_ACC_DOC_CREATE_ACKNOW	Accounting document acknowledgment	ZI_CONSTANTPARAMETERSF
ZCL_MB_CHECK_LINE_LOGIC	Line check	ZI_USR05
ZCL_DERIVE_PSA_MAIN	PSA derivation	ZDB_PSA_CCODE

Each of these performs multiple SELECTs per invocation (USOBX_C authorization checks, custom Z configuration tables, **I_PRODUCT** / **IPRODUCT** CDS views). Aggregated, this layer represents the largest single portion of the runtime.

2.2 Syclo mobile framework (~20% of the cost)

Approximately 30 **/SYCLO/*** tables are being queried per movement:

```
/SYCLO/CA002F, /SYCLO/CA002L, /SYCLO/CA002T
/SYCLO/CA003, /SYCLO/CA003M, /SYCLO/CA003D
/SYCLO/YCA002F, /SYCLO/YCA002L, /SYCLO/YCA002
/SYCLO/YCA003, /SYCLO/YCA003M, /SYCLO/YCA003D
/SYCLO/MDW00, /SYCLO/YCA000ST
```

Callers: **/SYCLO/CL_CORE_WD_TOOLS**, **/SYCLO/CL_CORE_BOM_EXCHOBJ**, **/SYCLO/CL_CORE_EXCH_SERV**, **/SYCLO/SAPLUPD_CORE**.

What it is doing: generating change pointers / exchange objects for a mobile application, on every goods movement. This is not required during an initial stock cutover.

2.3 CIF / APO synchronization active

Table hits include:

```
/SAPAPO/MARM, /SAPAPO/MATKEY, /SAPAPO/MATGROUP
/SAPAPO/MATLWH, /SAPAPO/MATTXT, /SAPAPO/MATPACK
/SAPAPO/MATEXEC
```

Callers: /SAPAPO/SAPLDM_MATERIAL, /SAPAPO/SAPLMAT_MASTER, /SCWM/CL_AF_MATERIAL_BASE_S4, /SCWM/CL_ERP_PI_S4.

CIF outbound integration to APO is active and firing per material touched.

2.4 Central Finance (CFIN) forwarding

- CFIN_MIG_LOG, FIN_CFIN_HIS_BCA
- CL_FIN_CFIN_CPAY_PAYT_CARD

CFIN is preparing / forwarding each FI document generated by the goods movement.

2.5 Joint Venture Accounting (JVA)

- Tables: T8J156X, IJVVALDINVGRP, I_JVAGLOBALCOMPANYPARAMETER, I_JNTVNTRVALDINVTGRP
- BAdI ZCL_JVA_FI_DOC_DFTCO_IMP invoked on virtually every document

2.6 Oil & Gas industry solution

- SAPLOIB_MM, SAPLOIB_QCI, SAPFOIH3
- HPM tables: MCHBH01, MARDH01, MCHB01, MARD01, ISEG01, ISEG02, OIH30H, OIB_DEF_INDEX_GM

Oil quantity conversion running per batch line.

2.7 Regulatory / Utility (FERC)

- /UTGFR/RS_FLAG, /UTGFR/T_EXTLDCO, /UTGFR/T_EXTCONF
- Regulatory accounting derivation via /UTGFR/CL_DERIVE_FERCACC per document

2.8 FI customizing consistency lookups

Top 3 internal data-access hits in `DataAccessInternal.csv`:

```
19,641 µs  CL_FINS_CUST_CONSISTENCY_DATA (Fetch TABL FETCH(Q))
16,987 µs  CL_FINS_CUST_CONSISTENCY_DATA
15,266 µs  CL_FINS_CUST_CONSISTENCY_DATA
```

Plus multiple **FINSC_***, **TCVPROF**, **TCVPROFT** lookups — configuration checks that should be table-buffered but are hitting the DB path repeatedly.

2.9 Material Ledger (ML) active per document

Tables: **CKMLKEV**, **CKMLCR**, **CKMLPP**, **CKMLPR**, **CKMLPRKEPH**, **CKMLPRKEKO**, **CKMLHD**, **CKMLO**, **CKMLMV001**, **CKMLMV007A** Callers: **SAPLMLCCS_CUSTOMIZING**, **SAPLMLCCS_DBACCESS**, **CL_FML_ACDOCA_M_EXTRACT_UTIL**

Cost component split executed per movement.

2.10 Number ranges not buffered

- **NRIV** accessed via **SAPLSNR3** (non-buffered path)
- Prepare / Open / Fetch cycles visible per document
- Should be running through the buffered path (**SAPLSNR1**) or with buffering enabled at SNRO level

2.11 RFC destination re-check per call

Tables read repeatedly via **CL_DEST_RFC** / **CL_DEST_ROOT**: **RFCDES**, **RFCATTRIB**, **RFCDESSECU**, **RFCDOC**, **RFCHECK**, **RFCCBWHITELIST**

2.12 Commit is not the bottleneck

Only **one Commit Work** event appears in the traced window (from **/SCWM/CL_MIG_STOCK**). Commit frequency is not the driver — the per-record customization and integration load is.

3. Recommendations — By Impact and Effort

Level 1 — Configuration Quick Wins (days, immediate gain)

#	Action	Where	Expected gain
1	Deactivate Syclo framework during cutover	Syclo customizing / IMG	-15 to -25%
2	Deactivate Z/Y BADIs (filter by migration user or IMG toggle)	SE18 / SE19 / IMG	-25 to -40%
3	Suspend CIF outbound to APO during load	CFC9 / CIF outbound queues	-8 to -12%
4	Suspend CFIN replication for the logical client during load	CFIN configuration / RFC destination	-3 to -6%
5	Enable buffering on MB and related number ranges	SNRO — turn off "no buffering"	-2 to -5%
6	Verify buffering on FI config tables (FINSC_* , TCVPROF , TCVPROFT)	SE13 (where technically allowed)	-2 to -4%

Realistic combined gain from Level 1: ~40–60% → 24h → ~10–14h, using configuration only, no development.

Level 2 — Design Change (weeks, larger gain)

#	Action	Gain vs. 24h
7	Replace <code>/SCWM/ISU</code> with a Z program calling <code>/SCWM/API_PHYSTOCK_CREATE</code> directly, with package size 100 + parallelism 5	~65–75% (24h → ~6–8h)
8	Migrate to Migration Cockpit object "Warehouse stock (EWM)"	~40–50%

Level 3 — Scope Reduction (days, potentially highest gain of all)

#	Action	Gain vs. 24h
9	Confirm whether all 28,687 records really need to be migrated (residual balances / neutral positions can be dropped)	10–100% depending on scope
10	Split the load by warehouse (partition per storage location / storage type) and run in parallel	additional -40 to -60%

4. What Will *Not* Help Significantly

Based on the trace evidence:

- **✗ Hardware upgrade** — the bottleneck is ABAP logic, not I/O.
- **✗ More work processes alone** — the serialization is inside each document.
- **✗ Individual SELECT tuning** — every single SELECT is fast; the cost comes from the **quantity** of them.
- **✗ HANA hints or new indexes** — the hot tables are already buffered or small.

5. Correction to the Original Migration Report

The original migration report (`Migration-BAPI-Report.md`, section **3.1 Inventory (EWM)**) assumed a maximum gain of **0–20%** because no direct BAPI exists for `/SCWM/ISU`.

The trace analysis proves this assumption is **too conservative**. A **60–75% reduction is achievable**, but through a different path:

- Not by replacing the API
- But by **eliminating the customization and integration layers** that fire per record during the cutover window

Most of the gain is **operational and cutover-configuration driven**, not development driven.

6. Suggested Next Steps

1. Validate with the functional teams which BAdIs are safe to deactivate during cutover (migration user flag).
2. Align with Basis on:
 - CIF outbound suspension window
 - CFIN replication pause

- Number range buffering
 - 3. Confirm with the mobile team whether the Syclo change pointers can be paused during cutover.
 - 4. Confirm scope of the 28,687 records — determine if any part is disposable.
 - 5. Prepare a small POC on a QA copy: run */SCWM/ISU* with the top customization layers disabled, on a controlled subset (e.g., 500 records), and measure the actual per-record time.
 - 6. Decide between Level 2 options (custom Z program vs. Migration Cockpit) based on the POC outcome.
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*Document generated on 2026-08-01 — ConcentoRDG Project — Trace analysis of */SCWM/ISU* (Inventory EWM)*